

# SINGLE BOARD COMPUTERS

## Are No Longer ‘Single’

Single board computers have quietly become much more than compact controllers. When a board now influences intelligence, longevity, and competitiveness, choosing one is no longer an engineering detail; it is product vision.

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Once chosen primarily for size and price, single board computers (SBCs) have become long-term platform decisions that directly define AI capability, system reliability, scalability, and product lifecycle. For today’s edge deployments, the SBC is no longer a peripheral choice; it is the architectural foundation on which intelligent systems are built.

The traditional view of an SBC as a compact controller handling basic I/O and automation tasks is now outdated. Modern SBCs are running AI inference at the edge, orchestrating multi-sensor data, driving multi-display HMIs, and operating reliably in harsh industrial and outdoor environments far from the cloud. As Venkatesh K., Sr Sales Manager (India), at ADLINK, explains, “SBCs are no longer just control platforms; they are designed as edge AI engines where inference happens locally rather than in the cloud, enabling real-time decision-making across industrial, transportation, and automation applications”.

From factory floors and railway yards to robots, EV chargers, kiosks, and smart cameras, SBCs have become the intelligence layer of modern edge systems. This transformation has not happened accidentally. Vendors are fundamentally rethinking how these boards are architected, thermally managed, expanded, secured, and supported over long deployment life cycles.

### SBCs are becoming AI-native, not AI-ready

The most significant shift reshaping SBC design today is the move towards edge-native AI. Applications such as vision-based inspection, object detection, anomaly detection, sensor fusion, predictive maintenance, and especially robotics cannot tolerate the lag or latency associated with cloud-dependent processing. These workloads must execute where data is generated—at the edge, in real time.

This requirement has driven SBCs away from purely CPU-centric architectures towards heterogeneous compute

### Key Drivers in the Indian Market

#### Smart Manufacturing & Industry 4.0

- Vision Inspection
- Predictive Maintenance
- Machine Monitoring



#### Transportation & Railways

- Wide-Temp Operation
- Vibration Resistant
- Long Lifecycle Support



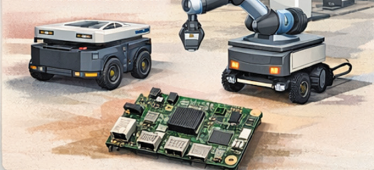
#### EV Charging & Smart Infrastructure

- Low-Power
- Secure Boot & TPM
- Remote Management



#### Robotics & AMRs

- Autonomous Robots
- Inspection Bots
- Collaborative Robots



#### Healthcare & Medical Electronics

- Diagnostic Equipment
- Imaging Systems
- Smart Monitoring Devices

